UNIT 5 · STANDARD 6.G.1

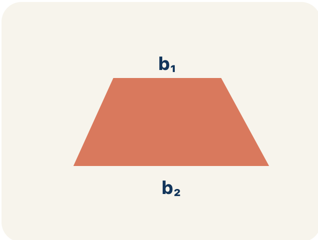
Area of Trapezoids

Lesson 5-2

Name: ____________ Date: ____________ Class: ____________

Key Vocabulary Level 2 Standard

Picture first, then the word, then a plain-language meaning. Say each word out loud.



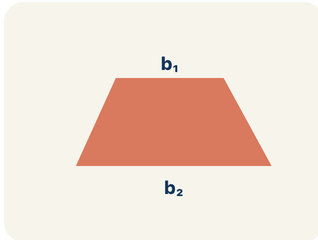
A table top wider than the bottom shelf — the top and bottom edges are parallel, the two sides slant inward

Trapezoid

Write the definition:

____

____



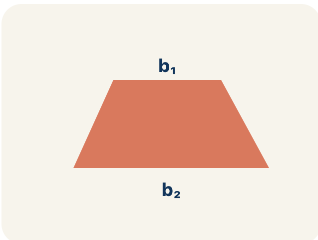
If the bottom of the trapezoid is 10 ft, then b1 = 10 ft in the formula A = ½(b1 + b2) × h

Base 1 (b1)

Write the definition:

____

____



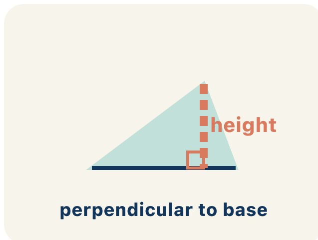
If the top of the trapezoid is 6 ft, then b2 = 6 ft; both bases are parallel to each other

Base 2 (b2)

Write the definition:

____

____



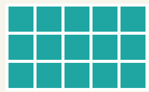
A dashed vertical line from the top base straight down to the bottom base at a 90° angle — NOT the slanted side

Height

Write the definition:

____

____



Area = 15 square units

A trapezoid with $b_1 = 10$, $b_2 = 6$, $h = 4$ has area = $\frac{1}{2}(10 + 6) \times 4 = 32$ sq units

Area

Write the definition:

Guided Notes `Level 2 Standard`

🎯 **WHAT WE'RE LEARNING TODAY**

I can find the area of a trapezoid using the formula A = ½(b1 + b2) × h.

✏️ **Fill in each blank as we go. Use the Word Bank to help you.**

📚 **WORD BANK — FILL EACH BLANK WITH THE BEST WORD**

Trapezoid | **Base 1 (b1)** | **Base 2 (b2)** | **Height** | **Area**

1. A quadrilateral with exactly one pair of parallel sides is a ______________.

2. One of the two parallel sides of a trapezoid is ______________.

3. The other parallel side of a trapezoid is ______________.

4. The perpendicular distance between the two parallel bases is the ______________.

5. The amount of flat space inside the trapezoid is its ______________.

🧮 Watch & Try — Worked Examples

See the notes in action: watch one worked all the way through, then try the next with the same steps.

👀 **I do — watch**

Follow each step as your teacher solves it.

Problem: What is the area of a trapezoid with bases 6 cm and 10 cm, and height 4 cm?

A. 32 sq cm

B. 64 sq cm

C. 24 sq cm

D. 40 sq cm

Step 1 $A = ½ \times (b1 + b2) \times h = ½ \times (6 + 10) \times 4 = ½ \times 16 \times 4 = 32$ sq cm.

✅ **Answer:** A. 32 sq cm

🤝 **We do — together**

Solve this one with your class using the same steps.

Problem: A trapezoid has an area of 45 sq ft, bases of 7 ft and 11 ft. What is the height?

A. 5 ft

B. 9 ft

C. 3 ft

D. 10 ft

Step 1 ____________________

Step 2 ____________________

Answer: ____________________

✏️ **You do — your turn**

Now try one on your own. Show every step.

Problem: What is the first step when finding the area of a trapezoid?

A. Add the two bases together

B. Multiply the two bases

C. Subtract the shorter base from the longer base

D. Divide the height by 2

Show your work:

Try It

Solve on your own. Check the answer key when you are done.

1. Blueprint 2: A trapezoidal skylight has bases of 9 ft and 7 ft and a height of 6 ft. What is its area?

A. 48 sq ft

B. 96 sq ft

C. 63 sq ft

D. 44 sq ft

Show your work:

____

____

____

2. Level 2 Extension: Two copies of a trapezoidal window (bases 5 ft and 11 ft) are flipped and joined to make a parallelogram with height 6 ft. The parallelogram's base is 5 + 11 = 16 ft, so its area is 96 sq ft. What is the area of ONE trapezoid window?

A. 48 sq ft

B. 96 sq ft

C. 16 sq ft

D. 192 sq ft

Show your work:

____

____

____

Stretch Your Thinking

Level 2 enrichment

Challenge task — explain your reasoning in full sentences.

A trapezoid and a parallelogram both have a height of 8 ft. The trapezoid has bases of 5 ft and 11 ft. The parallelogram has a base of 8 ft. Which shape has the greater area? Explain your reasoning using the formulas.

Sentence starter: The trapezoid's area is ____ because ½ × (____ + ____) × ____ = ____. The parallelogram's area is ____ because ____ × ____ = ____. The ____ has a greater area by ____ sq ft.

Show your work:

____

____

____

____

Reflect — Exit Ticket

A trapezoid has bases of 9 inches and 5 inches, and a height of 6 inches. What is its area?

A. 42 sq in
B. 84 sq in
C. 27 sq in
D. 42 in

Your answer:

____

____

____

Answer Key & Teacher Guide

1. **Notes 1:** Trapezoid
2. **Notes 2:** Base 1 ($b1$)
3. **Notes 3:** Base 2 ($b2$)
4. **Notes 4:** Height
5. **Notes 5:** Area
6. **We Try (notes):** A. 5 ft *— $A = ½ \times (b1 + b2) \times h \rightarrow 45 = ½ \times 18 \times h \rightarrow 45 = 9h \rightarrow h = 5$ ft.*
7. **You Try (notes):** A. Add the two bases together *— The formula is $A = ½ \times (b1 + b2) \times h$. The first step is to add the two parallel bases.*
8. **Try It 1:** A. 48 sq ft *— $A = ½ \times (9 + 7) \times 6 = ½ \times 16 \times 6 = ½ \times 96 = 48$ sq ft.*
9. **Try It 2:** A. 48 sq ft *— One trapezoid is half the parallelogram: $96 \div 2 = 48$ sq ft. This matches $A = ½ \times (5 + 11) \times 6 = ½ \times 16 \times 6 = 48$ sq ft — the ½ comes from the trapezoid being half of the parallelogram.*
10. **Exit Ticket:** A. 42 sq in *— $A = 1/2 x (b1 + b2) x h = 1/2 x (9 + 5) x 6 = 1/2 x 14 x 6 = 42$ square inches.*